

EEG Can Be Used to Measure Embodiment When Controlling a Walking Self-Avatar

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ABSTRACT

It has recently been shown that inducing the ownership illusion and then manipulating the movements of one's self-avatar can lead to compensatory motor control strategies in gait rehabilitation. In order to maximize this effect, there is a need for a method that measures, and monitors embodiment levels of participants immersed in VR to induce and maintain a strong ownership illusion. The objective of this study was to propose a novel approach to measuring embodiment by presenting visual feedback that conflicts with motor control to embodied subjects. Twenty healthy participants were recruited. During experimentations, participants wore an EEG cap and motion capture markers, with an avatar displayed in a HMD from a first-person perspective. They were cued to either perform, watch or imagine a single step forward or to initiate walking on the treadmill. For some of the trials, the avatar took a step with the contralateral limb or stopped walking before the participant stopped (modified feedback). Results show that subjective levels of embodiment correlate strongly with the difference in μ -ERS power over the motor and pre-motor cortex between the modified and non-modified feedback trials.

Keywords: Virtual reality, μ -rhythm EEG, event-related-potentials, gait rehabilitation, mirror neuron system.

1 INTRODUCTION

Virtual reality (VR) -based rehabilitation has seen an important gain in recent years, fueled by the availability of affordable mass-market VR devices and the several advantages such technology offers for patients and researchers [22,3]. One such advantage is the high level of control that can be exerted on all aspects of a participant's virtual environment (VE). When a participant is immersed in VR through the use of a head-mounted device (HMD), there is the added possibility of controlling his body self-representation in the form of a self-avatar.

There are examples in the literature where imitation of the movements of a simulated avatar or representation of movement by

an avatar led to motor improvements [17,46]. In 2014, Caudron and colleagues [6] showed that a simple avatar that replicates real-time anteroposterior trunk position and orientation of the head of patients with Parkinson's disease improved their postural balance. There are also randomized controlled trial studies that reported that controlling an avatar's gait in VR physically and mentally (through a BCI) could be beneficial for gait rehabilitation [8,13, 15,23,27,34,41-44,51] and for improving balance in chronic stroke patients [34]. Moreover, treadmills combined with VR scenarios have proved to be effective for post-stroke gait rehabilitation [13,44]. In their study, Rizo et al. [44] showed the advantages of using VR in gait rehabilitation by creating an obstacle avoidance VE system during walking in chronic post-stroke patients.

It has also been shown that in the earlier stage of pathology, motor imagery of the intended movements, combined with virtual feedback, can aid patients to gradually recover from impairment [8,23]. This technique was then applied for gait rehabilitation in stroke. For example, Kilicarslan and colleagues [27] pioneered the deployment of BCI systems to control lower-body powered robotic exoskeletons by subjects with a spinal cord injury.

Leeb and colleagues [31] reported on a 35 year old male tetraplegic subject, who learned to control a BCI where signals of imagined foot/right/left movements were used to control walk/turn-left/turn-right. He navigated in a VR scene, in order to move from avatar to avatar by movement imagination of his paralyzed feet. Concerning the experience with the interaction, he mentioned that "I thought that I was on the street, and I had the chance to walk up to the people".

These feelings described by the participant in such studies demonstrate that humans can be successfully embodied in a surrogate body, either of an avatar [2,49] or a robot [30]. Embodiment is the gradual process of the perceptual illusion that artificial body parts or full bodies can be perceived by people as their own [29]. Embodiment has three main components: body ownership, sense of localization and presence, and sense of agency and control. Agency is the feeling of authorship that we experience when initiating and controlling an action and distinguishes our own self-generated actions from those actions generated by others [4]. Agency requires the intention to carry out the action, and subsequently a match between its predicted and actual sensory consequence [4,9]. The sense of agency starts with the feeling of agency, which is triggered at the very early stages of the action. Then, once the feedback has been perceived and processed, the judgment of agency results from the computation of the comparison between the predicted and actual outcomes of the

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action [4]. The second embodiment component is the sense of presence and localization. Sense of presence is the psychophysiological state which reproduces realistic behaviors and physiological responses as if the subject was experiencing a real-life situation [10]. The third embodiment component is the sense of body ownership and mineness over a static manikin body that substitutes for the real body, which was first shown by Petkova and Ehrsson [40]. It is the feeling that is described in statements such as “This is ‘my’ hand,” and occurs when the visual and tactile information coming from this object spatiotemporally correlates [4]. Evidence suggests that first person perspective over the virtual body can be a sufficient condition to create the sense of body ownership and presence [35], but that synchronous movement between the real and virtual body can also contribute strongly to the sense of embodiment by inducing a strong sense of agency and control in the virtual reality environment [2,18,39].

However, is there a way to determine if someone has reached the sense of embodiment in the context of its three components? And to what degree? Questionnaires are currently the most used method to assess the different dimensions of embodiment [19,24]. This method of assessment has the obvious disadvantage of being a subjective evaluation that is dependent on a participant's interpretation of the different questions. Moreover, questionnaires do not enable real-time/in-task recordings of the level of embodiment [24]. Therefore, researchers have combined neurophysiological measurements [7, 9]. Most of these studies used positron emission tomography (PET) or fMRI but these methods are not suitable for everyday monitoring of embodiment in VR experiments because they are neither portable nor inexpensive. This is why recently some researchers have started to use electroencephalography (EEG) and near-infrared spectroscopy (NIRS) instead.

For example, Sanchez-Vives and colleagues [47] found that participants retrieved their physical hands when their virtual hand were threatened with a knife, resulting of a brain activation of μ -ERD (8-12 Hz) over the motor cortex and central-parietal areas of the brain, which suggests a potentially new measure of virtual embodiment. Clemente et al. [7] compared EEG presence during observation and control of navigation VE. They found an increase of frontal theta and alpha activity during the perception of presence. In normal circumstances, when our ongoing actions and the predicted sensory consequences of these actions (feedback) are coherent, we experience the sensation of agency with respect to our actions (“this action is mine”), and we are typically not even aware of such considerations [11,14]. However, in the case where there is a conflict between the predicted consequences of our actions and their actual consequences [20,48], we might detect an agency violation through an error detection mechanism. This mechanism might be constantly checking whether the final sensory feedback is coherent with expected sensory consequences of our actions, created using an internal copy of our motor commands. These sensory feedback estimations during movement may rely strongly on previous representations of the body in terms of limb position, movement, or posture which normally give us a natural sense of being the agents of our actions [16,21,50]. Padrao and colleagues [38] investigated the neurophysiological correlates of modified feedback and found a parietal N400 elicited by error monitoring loops after such violation, which typically characterizes semantic or conceptual violations. They also found that the amplitude of the

N400 correlated with the subjective feeling of body-ownership. When the same participants merely observed the avatar correct and error movements, no parietal N400 was elicited. This is what [32] confirmed, when participants looked at pictures representing themselves or other, and found that parietal positive ERP component P200 was less in the self-pictures.

Another explanation of this activity is the mirror neurons system that lies in the parietal and pre-motor cortex [36]. A mirror neuron is a neuron that fires both when someone acts or observes the same action performed by another [45,26]. Thus, the neuron “mirrors” the behavior of the other, as though the observer were itself acting, which helps in understanding the actions of other people, and for learning new skills by imitation.

But is there any relationship between the effectiveness of any training using the VR, and the level of embodiment in the VR? Alimardani and colleagues [1] showed in the case of BCI-control for a moving avatar or robot, the arousal of embodiment for the user is assumed to promote his involvement in the motor imagery task and enhance his skills in the navigation and operation. The effectiveness of controlling a VR is linked to the degree of embodiment, so the more the participant feels ownership, the more effective the results are [44]. For instance, in the case of amputees with a neuro-prosthetic limb, the long term and efficient usage of the limb depends on how well the patients accept the limb as an integrated part of their own body rather than a tool attached to them. The main objective of this study was to propose a novel approach to measure virtual embodiment during gait using EEG.

2 MATERIALS AND METHODS

2.1 Participants

Twenty neurologically healthy naive participants (13 women, 7 men; aged 23.3 ± 3.93 years old) volunteered to take part in this study. They were recruited through University electronic message boards. The inclusion criteria were that participants be 18-35 years old, in good general health, with good vision (with or without glasses), not taking any medication that acted on the central nervous system and not suffer from motion sickness.

2.2 Protocol and Experiment Design

This experiment consisted of 3 conditions where the participants were asked to “do”, “imagine” or “observe” three different tasks while looking at the lower limbs of their self-avatar through a head-mounted display (HMD). The duration was approximately 3 hours, including preparation time and regular breaks. The experiment was approved by the Research Ethics Committees of the University of Montreal Hospital Center (CHUM) and of Ecole de technologie superieure (ETS), project ID number 16.170.

In the “do” condition, brain activity was recorded while the participants were physically controlling the movements of the avatar in real time. The main goal for this condition was to measure embodiment during the physical gait control in an immersive virtual reality environment. In the “imagine” condition, brain activity was recorded while the participants were imagining the avatar moving, without the physically moving. The main goal of this condition was to measure embodiment during motor imagery of moving in an immersive virtual reality environment.

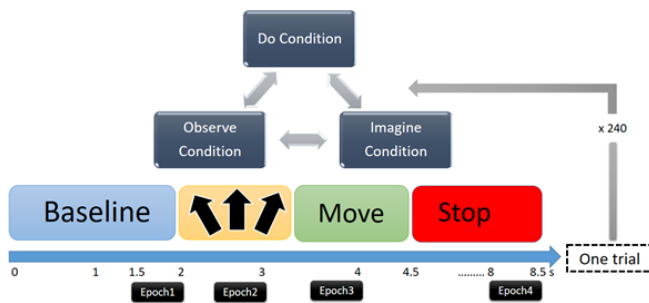


Figure 1: Experimental design of the study. Above are the conditions of the experiment, in the middle is the design of one trial. Below are the epochs (time slots) used in the analysis.

In the “observe” condition 3, brain activity was recorded while the participants were only observing an avatar moving, without physically moving. This condition was to measure embodiment during observing the gait of an avatar, and to rule out the possibility that any brain activity occurring during the “do” or “imagine” conditions was due to mere observation.

The experimental design is shown in Figure 1. Each condition consisted of 240 trials. Each trial was 8.5 seconds long and started with a two-second pause in order to acquire a baseline EEG recording. An arrow was then presented on the virtual floor in front of the participant for a period of 1.25 seconds. This arrow either pointed to the left, cueing a step with the left foot; to the right, cueing a step with the right foot; or forward, cueing to start walking. Upon receiving the cue, participants either performed, imagined or observed the appropriate action, depending on the condition they were in. They did this while standing on an instrumented treadmill.

For the first 60 trials of each condition, participants received concurring visual feedback in the form of the avatar performing the cued action (non-modified feedback - NMF). The avatar performed the action in real time for the “do” condition or 1.25 seconds after the cue onset for the “imagine” and “observe” conditions. These first 60 trials were to engage the participant in the experiment and to create the sense of embodiment. In trials 61 to 240, for one out of every 5 trials, the avatar provided conflicting visual feedback (modified feedback - MF). For these trials, when the cue was for a single step, the avatar took that step with the contralateral limb. When the cue was to start walking, the avatar started walking but stopped walking before the end of the trial, while the participant was still walking or imagining himself walking. This movement violation was implemented to create a mismatch between the intended and resulting movement and measure for an EEG-response to the modified feedback that is dependent on being embodied.

EEG data was only analyzed offline. The feedback provided to the participants was therefore not dependant on their brain activity.

2.3 Experimental Setup

Participants were immersed in a VE using an Oculus Rift (Consumer Version 1) HMD. The VE was developed in Unity 3D game engine and consisted of a virtual hallway (Figure 2), which was a replica of the hallway in our research center, although the virtual hallway was designed to be infinite. Horizontal lines were added to the floor design in order to for forward movement to be more easily perceived while looking at one’s feet. A virtual self-avatar was displayed from a 1st person perspective with respect to

the participant. The action cues that were presented in front of the virtual feet consisted of a yellow arrow pointing left, right or straight ahead. When the avatar took a step or started walking, he moved forward in the virtual hallway, thus resulting in a realistic optical flow of the VE.



Figure 2: A 3PP view of the virtual avatar standing in the middle of the VE (left image); a 1PP view of what the participants saw in the HMD, showing the cue for a step with the right foot (right image).

For the “do” condition, a set of 15 rigid bodies containing reflective motion-capture markers were placed on the participants’ bodies (Figure 3). A 12-camera Vicon optoelectronic motion-capture system with Vicon Tracker software was used to track the participants’ movements. During a short calibration phase, the participant performed a series of squats, hip circumduction movements and upper limb rotations in order to identify the position of their joint centers to align them with those of the virtual avatar’s rig.

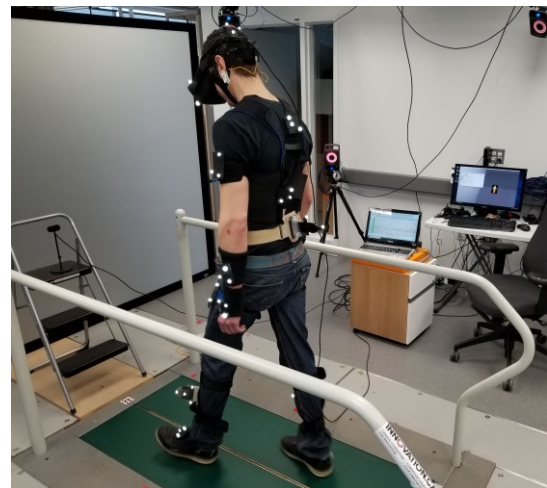


Figure 3: The experimental set-up. A participant takes a step on the treadmill while wearing the mocap markers, HMD and EEG cap

Movements were then applied to the virtual avatar in real-time, via TCI-IP protocol. The movements of the avatar had no noticeable delays when compared to the participant's real movements.

The EEG was recorded using a Smart BCI system consisting of a cap with 19 Ag/Cl cup electrodes set according to the 10-20 system and fixed to the scalp with conductive gel (SignaGel). These electrodes were grounded to AFz and referenced to both ears using an ear clip on each ear. The electrode positions were configured to accommodate this study (Figure 4), covering the

whole scalp with a higher density above the pre-motor, motor and parietal areas, which were the main regions of interest. The EEG device sends EEG data via Bluetooth to the EEGStudio software, which in turn streams them in real time via API to Matlab.

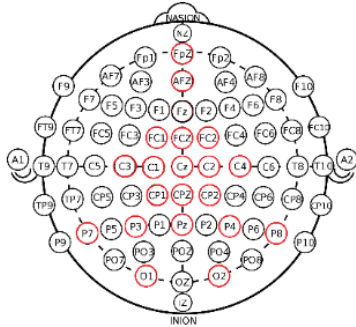


Figure 4: The EEG-electrode configuration for the current study. Red circles indicate the positions of electrodes.

2.4 EEG Data Pre-Processing

The data were recorded using EEGStudio, pre-processed and processed using Matlab/EEGLab. EEG signals were band-pass filtered in 4-30Hz, then artefact removal was performed using 3 methods: 1- a semi-automatic method, where epochs containing abnormal values (inferior to $-200\mu V$ or superior to $200\mu V$) or abnormal trends (maximal slope superior to $200\mu V$ with a R-squared limit fixed to 0.3) were rejected; 2- an approach based on blind source separation (BSS) algorithms was used, which includes an automated independent component analysis (ICA) to isolate and remove electro-ocular components from the EEG data [25]; 3- with visual inspection, artefacted epochs were also rejected. Data were then re-referenced using Common average referencing (CAR). A baseline removal algorithm was applied, using the 200ms preceding the cueing (apparition of the arrow) as the baseline. Data were then epoched into 4 main 500ms epochs: preceding the cueing, following the cueing, during the movement and after the end of the movement. The rationale behind this epoching was that the feeling of agency is supposed to occur during the trial beginning period while the judgment of agency should occur during the feedback processing period. Finally, each epoch was filtered into 2 bands: alpha band (8-12 Hz) and SMR (12-15 Hz).

2.5 EEG Data Offline Analysis

In the above-mentioned epochs and frequency bands, spectral power analyses were performed. A statistical analysis (paired t-test) was then performed in order to determine if the recorded signal power was significantly different over all of the above-mentioned cases. These analyses were performed using permutation statistics, with a threshold p-value fixed at 0.05, completed by a false discovery rate (FDR) correction for multiple comparisons.

2.6 Questionnaires

After each condition, participants were asked to answer a questionnaire consisting of 9 questions with 7-point Likert scales. These questions were related to the sense of presence (1-3), body ownership (4-6) and sense of agency and control (7-9) [38,29,33]. It took approximately one minute to complete the questionnaire.

3 RESULTS

3.1 Neurophysiological Results

Data analysis revealed significant differences between trials with modified and non-modified visual feedback only in μ frequency band. All results presented in this section are therefore within this frequency band. Brain activity related to the left- and right-footed steps was found to be laterally inverted, so the results were mirrored and combined in one plot in order to facilitate the analyses. These results are depicted in Figure 5.

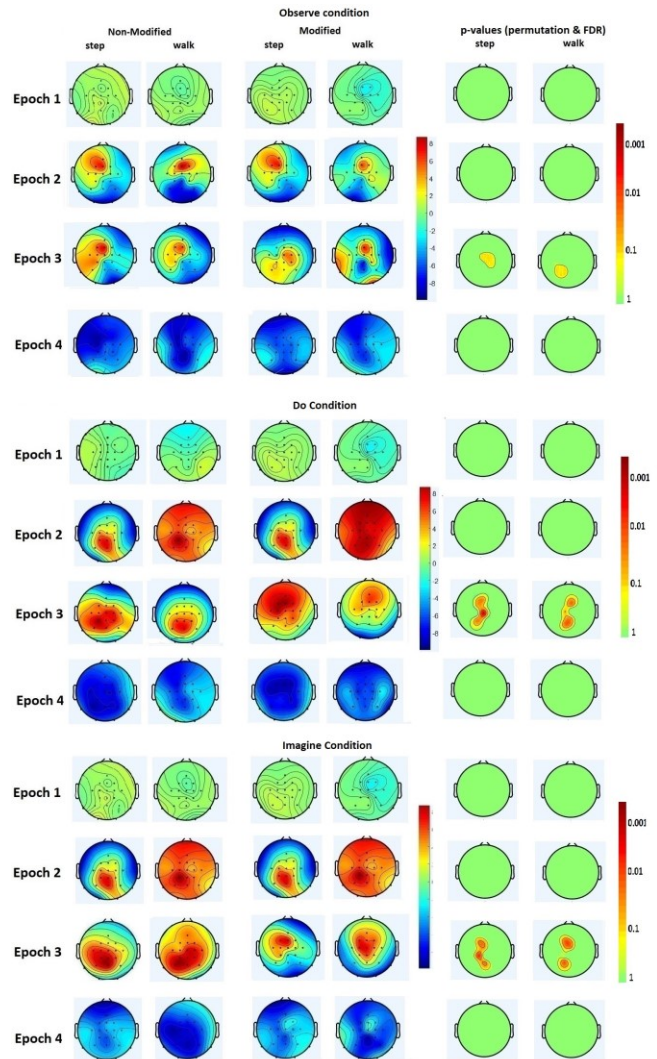


Figure 5: Spectral power maps of the μ frequency band (8-12 Hz) for the different epochs (time slots), conditions and cases. The analysis shows when controlling the avatar's gait, a strong central and parietal ERS in the case of non-modified feedback, versus a stronger frontal ERS in the case of modified feedback.

In the "do" and "imagine" conditions, the spectral power peaks over the central and central-parietal areas of the brain, with a stronger activation of the central-frontal areas in the walk stimuli in epoch 2 (500 – 1000 ms after the cueing appeared). These brain activations are a representation of a larger event-related synchronisation (ERS - i.e., increase of the signal power compared to baseline) and they may be specific to the higher feeling of

agency. These spectral power increases occurred with no significant difference between the “do” and “imagine” conditions ($p=0.2$) but were significantly weaker in the “observe” condition, for both central-parietal ($p=0.04$ and $p=0.001$) and central-frontal ($p=0.04$ and $p=0.001$).

For NMF trials of the “do” and “imagine” conditions, the high central-parietal and central ERS seen in epoch 2 remained in epoch 3 (250 – 750 ms after the visual feedback started) on the central-parietal ($p=0.3$) but spanned to left-parietal areas of the brain ($p<0.05$). Moreover, there was no significant difference in central-parietal area between the “do” and “imagine” conditions ($p=0.3$). In the “observe” condition, the analysis of NMF trials did not show the left-parietal ERS component observed during the “do” and “imagine” conditions ($p<0.001$).

In contrast, for MF trials (i.e. in low agency) the “do” and “imagine” conditions show that the high ERS was diminished or disappeared from the central-parietal region ($p<0.001$) and became more central-frontal after a walk stimulus ($p<0.05$) and left-central after the step stimuli ($p<0.001$). Moreover, there was significant difference in the epoch 3 central-parietal ERS between the MF and NMF for the “do” ($p<0.05$) and “imagine” ($p<0.001$) conditions.

Pairwise comparisons showed that the central-frontal ERS amplitude was significantly increased for the “do” ($p<0.05$) and “imagine” ($p<0.001$) conditions, when compared to the “observe” condition. No significant differences were found between MF and NMF trials for the “observe” condition.

3.2 Behavioral Results

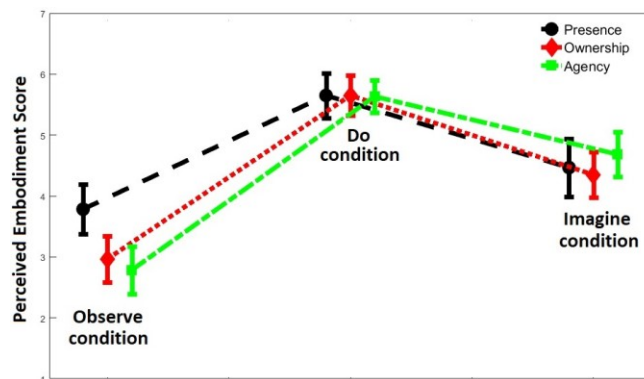


Figure 6 Behavioral results based on the questionnaire answers obtained after every block and decomposed into the 3 embodiment components. The score for each component was computed by averaging the three questions related to this component

In the questionnaires, there were 3 questions to rate each of the components of embodiment. The score for each component was computed by averaging the three questions. A high score on the component average indicates a stronger sensation for this embodiment component (Figure 6). The Shapiro-Wilk test revealed that the normality assumption is rejected, so the Wilcoxon test was performed over all paired samples of the questionnaire responses.

Overall, these results are consistent with our expectations and show that in general, when physically controlling the avatar (“do” condition), the perceived embodiment is higher than when mentally controlling the avatar (“imagine” condition), which is in turn higher than when observing the gait of an avatar.

Within the “do” condition, the three embodiment components had almost the same perceived levels (at 80%), with no significant differences. This suggests that the modified feedback influenced the embodiment components altogether. However, in “imagine” and “observe” conditions, we can see a relatively higher score of sense of presence, compared to the two other components of embodiment. In the “observe” condition, the score for the sense of presence was significantly higher than the scores for agency ($Z= -2.3532$, $p=0.018$) and ownership ($Z= 1.8716$, $p= 0.06$) but also significantly lower than the presence score for the “do” condition ($Z= -3.8701$, $p<0.001$).

In the “imagine” condition, the sense of agency score was significantly higher than in the “observe” condition ($Z= -4.0860$, $p<0.001$) and significantly lower than in the “do” condition ($Z= 2.8801$, $p=0.004$).

A correlation analysis (Wilcoxon rank sum test correlation) was performed between the embodiment questionnaire results and the brain activity of the central-frontal (electrodes FCz, FC1 and FC2) and parietal areas (electrodes CPz, CP1, CP2, Pz, P3 and P4). This brain activity was represented as the difference in mean amplitude of the ERS between MF and NMF trials, elicited at epoch 3.

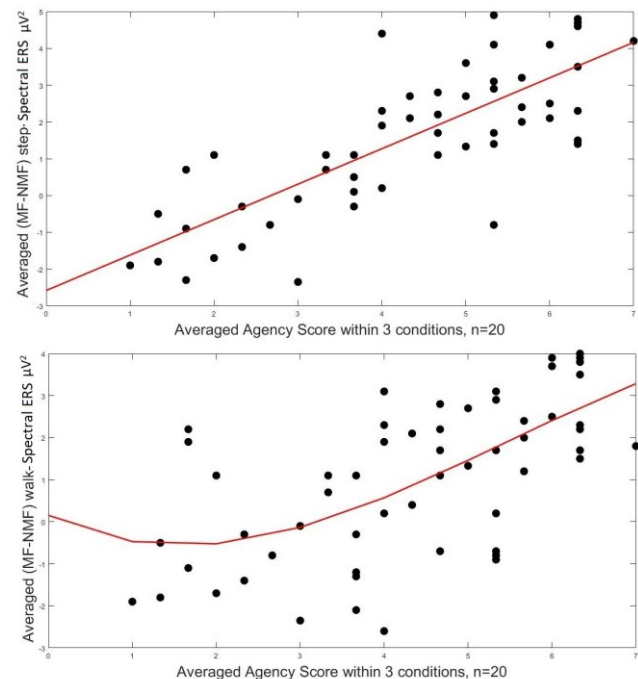


Figure 7 A correlation analysis over the subjective strength of agency, and the mean amplitude of the epoch 3 ERS over the 3 different conditions. This amplitude was computed subtracting the steps/walk NMF-ERS from the steps/walk MF-ERS over the parietal and central frontal electrodes

In other words, it is a correlation between the subjective evaluation of embodiment and the mean effect of the MF over the central-frontal and parietal areas. This was done for each of the 3 conditions. Results show a positive correlation between these measures ($r_{\text{step}}= 0.69$, $r_{\text{walk}}=0.61$). When using only the scores of the sense of agency, the correlation is stronger ($r_{\text{step}}= 0.73$, $r_{\text{walk}}=0.61$). This result, shown in Figure 7, suggests that

participants who experienced stronger embodiment, elicited stronger ERS modulations in response to agency violations.

4 DISCUSSION

This study focused on the possibility of measuring the perceived feeling of embodiment using EEG, when a user controls an avatar that represents his body and mimics his movements, or his imagined movements, in real time. To do so, an experiment investigated the effect of providing modified visual feedback to embodied participants, in the form of a self-avatar whose movements were incongruent with those performed or imagined by the participant (modified feedback).

When delivering modified feedback (low agency), a strong and long central-frontal ERS was found in the “do” and “imagine”, representing the judgment of agency and resulting from the error-monitoring loops and the complex compensatory cognitive control mechanisms that were triggered after the avatar error, where the user intends to stop the motion on reaching the goal position. This activity was found to be much lower in the absence of modified feedback, and this is may be due to the normal activity of the error-monitoring neuronal circuits in the brain area.

Our ability to recognize ourselves as agents of our own behavior depends on constantly monitoring the sensory consequences of our ongoing actions. When a mismatch is detected between any of these internal predictions and reafferent signals, a violation of the sense of agency might be triggered. Thus, this frontal-central ERS could be reflecting the output of this comparison process, which might lie at the core of the movements monitoring loop. This pattern of greater MF ERS over the central-frontal areas may be specific to the judgment of agency, where the user intends to stop the motion of the in order to reach the target position [37]. This is due to the mirror neurons system in the frontal lobe, firing neuronal activity in order to understand and manage this erroneous action of the avatar. Moreover, the ERS over the parietal areas may be due to the role of this part the brain in spatial navigation when the feeling of agency occurs.

These neurophysiological measures revealed the same patterns as the ones described in the literature:

- 1- A strong central-frontal and central-parietal μ -ERS was revealed in the non-modified feedback epochs (high sense of agency).
- 2- This parietal μ -ERS was due to role of the angular gyrus in the inferior parietal cortex to the feeling of agency [12] and the comparison processes between predicted and actual consequences of ongoing actions [5,28].
- 3- A strong central-frontal and left-frontal μ -ERS was revealed in the modified feedback epochs (low sense of agency).

Remarkably, the step MF-ERS over the left central area was higher than the walk MF-ERS for the “do” and “imagine” conditions. This may be due to the complex compensatory cognitive control mechanisms that were triggered after the avatar moved incongruently with regards to the participant. When physically or mentally controlling the gait of an avatar, the movement violation of the walk (walk MF) was presented by delivering an absence of feedback. However, the movement violation of the steps (steps MF) was presented by delivering an incongruent feedback, and the latter seems to be more power-consuming by the brain than the former. This is aligned with previous findings, that the cognitive

return of an anti-saccadic movement is stronger than the cognitive return of a movement inhibition.

In contrast, when observing the gait of an avatar in that immersive virtual reality environment, no central-frontal or parietal μ -ERS changes occurred after delivering modified feedback, but only few parietal and frontal ERS traces were found. This was confirmed by the scores in response to questions such as ‘It felt as if I was in a corridor’ or ‘There were times when, I forgot my presence in the real world, believing that the avatar was me’, where we can see a relatively high score of sense of presence and localization,

This may be due to the immersive virtual reality environment, and the avatar that was displayed in 1PP. This result is consistent with previous findings, that watching an avatar’s gait in a 3D immersive virtual environment is good enough to create the sense of embodiment. On the other hand, could the modified feedback influence the low scores of the two other components in the watching block? Actually, and as stated above, the modified feedback influences all the embodiment components, thus the high score of the sense of presence rules out the possibility of effect of modified feedback over the watching condition.

In this study, a significant correlation was observed between the amplitude of the frontal-central and parietal ERS component and the subjective feeling of body ownership. The greater the subjective feeling of body ownership, the stronger the ERS amplitude or the electrophysiological signature of agency violation. More importantly, this activity was not found when observing the modified feedback of an avatar’s gait. In this condition, the participants’ questionnaire scores showed that they felt immersed but felt significantly less in control, which supports the theory that the strength of frontal-central ERS can be used to measure the sense of agency over the gait of a virtual avatar when delivering modified feedback.

Overall, the “imagine” condition induced a high level of embodiment toward the self-represented avatar (as measured by body ownership, localization, and agency) [4,20,39]. Even though participants did notice that they could not always control the avatar movements, as reflected by the scores in response to questions such as ‘The movements of the avatar corresponded to my movements / imagined movement in real time’ or ‘There were times when I felt that I was walking with my walk and not with someone else’s walk’, their sense of agency score was higher than in the “observe” condition and lower than the “do” condition. To summarise, these analyses suggest that when physically or mentally controlling the gait of an avatar, the three embodiment components can be measured by neuro-markers represented by the central-frontal and central-parietal μ -ERS changes after modified feedback, potentially reflecting the sense of presence, the sense of agency with its two components (the feeling and the judgment of agency) and the sense of ownership.

5 CONCLUSIONS

This study showed that it is possible to use EEG to measure the level of embodiment when physically or mentally controlling the gait of an avatar, through the neuro-markers elicited after providing modified feedback of the avatar’s gait. To our knowledge, this is the first study to show an EEG response to incongruent visual feedback of a self-avatar during gait. It is also the first the show a correlation between this EEG response and subjective

questionnaires of embodiment of an avatar, in the context of lower limb-movements.

To conclude, our results have important implications for the development of a more objective method of assessing the sense of embodiment, based on physiological data. Most importantly, the approach used in this study could eventually be used for online real-time monitoring of the sense of embodiment. This could ensure embodiment is maintained in order to maximize the benefits of rehabilitation protocols in embodied immersive VR.

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